Securing Big Data: A Multi-Layered Approach

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Abstract. Traditional big data technologies have made it possible to collect, process and analyze data from new sources, the security of which needs to be assured. The recent high demand for scalable data-centric applications in areas such as government, healthcare, finance, and e-commerce has increased the number of systems that face potential security threats due to their immense data volume, velocity, and variety, which cannot be handled by traditional security solutions. This paper presents a holistic security assurance approach for big data contexts, as well as an automated compliance module and several other novel techniques, including multi-layer encryption, attribute- based access control (ABAC), machine learning-based intrusion detection system (IDS), and blockchain-based audit logs. The results of the experimental evaluation demonstrate that the proposed approach can effectively improve data confidentiality, integrity, and availability (CIA) while preserving an acceptable level of performance.

Keywords: Big Data, Security Assurance, Data Privacy, Threat Analysis, Regulatory Compliance, Hadoop Security, Cloud Security

1. Introduction

Big data have been a must-have to any digitalization and transformation. It is an essential element in e-commerce, healthcare, smart cities, and financial services. It’s easy to imagine the tremendous amount of real-time data that the stock market generates. Data from the stock market exist in the form of market feeds, trading patterns and transactions. These data are then consumed by investors, regulators and companies to make their own investment decisions. Financial big data are also highly sensitive, given its high volume, volatility and direct impact on the financial bottom line.

1.1 Motivation

Successful functioning of modern economies depends on the availability of reliable and timely financial data. For this reason, the security of information on the stock market is an issue of crucial importance. A single security incident is able to overturn the regulatory status quo, shake investor confidence and compromise the stability of financial institutions. The motivation for this research is the immediate need to provide timely and reliable security assurance to protect the confidentiality of information, prevent loss of integrity due to malicious intent and ensure the availability for prompt and legitimate analysis. In order to meet this need, this paper has combined several defensive techniques, such as AES-256 encryption, SHA-256 hashing to verify data, secure API / SSH communication channels and role-based access controls in current compliance requirements to propose a practical and auditable, robust protection layer for real-time financial big data.

1.2 Main contribution

(a) An analysis of the current literature shows that although there have been developments in encryption, access control, anomaly detection, and compliance, the majority of the existing solutions provide these as stand-alone.
(b) Very few of the proposed frameworks actually provide encryption, access control, compliance, and anomaly detection and detection as a joint security assurance measure for financial big data.
(c) Stock market datasets in particular are often fast-moving, high-volume, and highly sensitive, necessitating the need for real-time protection as well as regulation compliance.
(d) An innovative approach was developed that combined AES-256 encryption, SHA-256 hashing, SSH-secured transfer, access control, and compliance enforcement for end-to-end financial dataset protection.

2. Literature Review

The area of big data security has already been covered in a previous document. The research in this field has given us an answer to the security problems raised by the processing of a large amount of data and the concerns which accompany them. We have access to state-of-the-art cryptography and access control systems, to strategies of compliance and several other security measures. However, the existing material is still not enough to provide a ready-to-use solution for the more specific and more sensitive/biggest challenge data. Thus, a comprehensive, yet flexible solution for implementation in the financial world of big data is required.

2.1 Encryption Approaches

Encryption was reported to be one of the primary methods to maintain big data securely in the major portion of study. A part of the study suggested the idea of homomorphic encryption where mathematical operation can be executed on encrypted data without decrypting it. In real-time financial transactions and environment, it is not easy to apply homomorphic encryption because it demands a very heavy load [1]. Another research represented the problem of secrecy and privacy in a big data and cloud environment. By using hybrid techniques, such as RSA and AES, security can be increased to an appreciable level. These hybrid methods are difficult to be applied because they increase the complexity [2]. However, AES-256 is widely popular because of its efficient trade-off between the efficiency of its algorithm and security. This is the reason why we consider AES-256 encryption to be used for our proposed model to perform massive financial data, which is rapidly increasing these days [3].

2.2 Access Control Mechanisms

Access control has also been addressed. Subashini and Kavitha [4] discussed Role-Based Access Control (RBAC), which is popular and widely used but lacks the flexibility required in big data and dynamic cloud environments. To overcome this issue, Xiao and Xiao [5] investigated Attribute- Based Access Control (ABAC), which offers fine-grained authorization by considering user attributes. However, research shows that RBAC and ABAC where system is still vulnerable to insider attacks and improper access, a problem or financial datasets requiring accountability and traceability [7], [9].

2.3 Intrusion Detection

Intrusion detection is another important application area in which the power of machine learning and anomaly detection techniques was misused. The use of classifiers to identify malicious activity based on system and data features was examined by authors [6]. Such a system may be helpful in the context of the stock market since it can identify unusual trading activity or unauthorized access attempts. Nevertheless, these methods are frequently retrospective and necessitate high-quality training data. They cannot, in and by themselves, provide strong guarantees about data confidentiality or integrity unless coupled with cryptographic protection mechanisms [8], [10].

2.4 Compliance Models

Regulatory compliance is another key big data security principle. The General Data Protection Regulation (GDPR), Health Insurance Portability and Accountability Act (HIPAA), and ISO/IEC 27001 information security standards all focus on accountability, data minimization and lawful processing. However, as authors noted, many compliance frameworks remain theoretical and do not provide technical enforcement mechanisms including encryption or integrity checks. This gap reduces their practical effectiveness in real-world financial systems [11], [12].

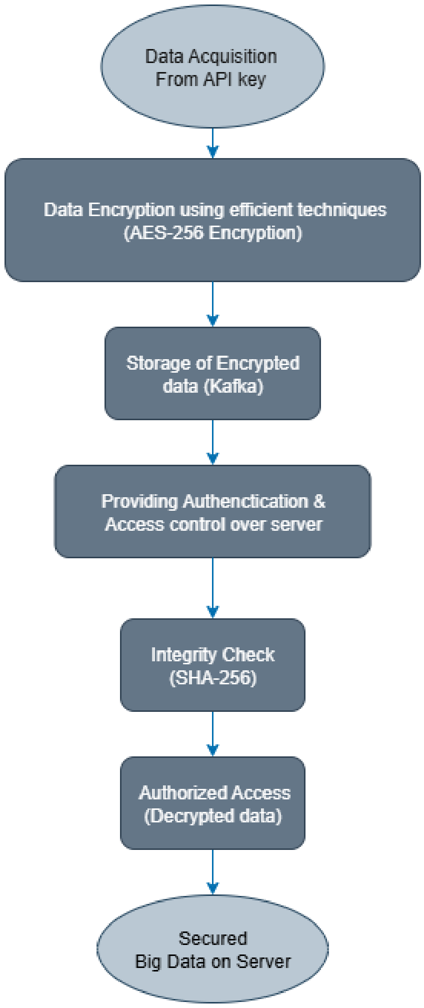

Figure 1. Overview of the Security Assurance Framework for Financial Big Data.

3. Architecture of Proposed Methodology

Figure 1 shows the framework of Security Assurance for Financial Big Data. The framework includes all the essential features and the interactive relationship between them. Figure 1 displays the flowchart showing how the confidentiality, integrity, and compliance with laws and regulations is sustained in the financial big data life cycle.

3.1 Enabling Technologies

The given framework secures financial big data. The pro- posed approach was developed over the real-time stock market information. As it is supposed to handle sensitive financial information securely and work with high-volumes of stock market data, the system needs to be endowed with a range of specific features which work jointly and help satisfy the requirements such as Confidentiality, Integrity, and Availability (CIA). In this framework several complementary enabling technologies work hand in hand, which not only makes the system robust, but also prevents vulnerabilities. However, each of the security technologies was selected after giving emphasis to both, its security strength and its implementation for the given purpose.

(a) Data Acquisition: The first step in this framework, where in the real-time stock market data is acquired from the generated API Key, and thus providing security to the data itself, so that it remains in a secure form and cannot be tampered by anyone or any unauthorized person to inject any false or manipulated data. The data is continuously fetched from the API as per the updation time frame, for this framework it is at the end-of-day.

(b) Data Encryption: The data encryption system proposed for the framework is AES- 256 encryption, this serves as the foundation of this framework. The AES is a symmetric key algorithm thus, this framework ensures strict confidentiality by encrypting the stock market records in both storage and transmission of data. Thus, even if attackers have unauthorized access to the storage system or intercepted data streams, the information remains indecipherable without the key. The AES-256 hash also provides the computational speed needed to support the high-throughput nature of financial data, which makes it a practical choice for big data in the financial domain.

(c) Kafka for Storage: Apache Kafka is the backbone for the stock market feed streams. The important thing in the case of financial systems, is that the data has to be ingested as close to real time as possible for accurate decision making. Kafka gives a low latency, fault-tolerant and highly scalable messaging streaming. Kafka allows stock data from an API feed to be pulled in, distributed, and consumed efficiently. It is a distributed system that is designed for high availability with the lowest possible loss, making it the ideal backbone for stock market data storage in the case of this framework. Kafka thus transports the encrypted data from stock markets in a secure manner to the next step in the case of this framework.

(d) SHA-256 Hashing: Protecting the integrity of the data, is just as crucial as protecting its confidentiality. The framework also, hence it uses SHA-256 hashing for added security. The unique cryptographic hash is created for every dataset collected and stored. The system recalculates the hash when the data is pulled up at a later time and compares it to its original hash value. In the case of a discrepancy, even if the smallest change has been made, which triggers a mismatch the system sends a prompt, thus detecting tampering. This process ensures that financial analysts and automated systems are always working with accurate and trustworthy data, which is crucial in the stock market system because even small changes in a few decimals can have an impact on analytics or even cause skewed investment strategies and lost opportunities.

(e) Secure Shell (SSH): Stock market data uses API keys that return the JSON objects encrypted through SSH as a protocol to secure the connections during data transmission through its network of various nodes and storage points to ensure data remains encrypted during its transit or transfers. It also prevents data tampering or third-party interference during transit or transmission such as man-in-the-middle- attacks (MITM). The approach provides enhanced confidentiality to record transfers by preserving secure socket layer (SSL) and transport layer security (TLS) in transmissions over the network for sensitive market data, to manage risks and improve accessibility to preserved records.

(f) Data Decryption: Encrypted stock market data is inaccessible until safely delivered to the proper receiver. However, only those with authorized clearance or approval are able to decrypt or see the real value stored or encoded in these records.

(g) Secured Big Data on Server: The repository server for stock market information is secured to ensure data from the collected records remain private and prevented from malicious attack. The collected data records are encrypted in-transit and at-rest to prevent them from becoming available as raw and unprocessed information by third parties with unauthorized access or clearances. Integrity also prevents the processed stock market data records from being easily modified or altered to ensure these records are also available and reliable for future access.

3.2 Models Used

To protect sensitive stock market information, the system follows a multi-layered security approach that ensures confidentiality, integrity, and controlled access.

(a) AES-256 Encryption: AES-256 is used to encrypt financial data, including market reports and trades. This algorithm transforms data into an unintelligible format that requires an authorized key to decrypt. It is quick enough to enable real-time processing of massive amounts of stock market data and incredibly resilient to brute-force attacks.

(b) SHA-256 Hashing: SHA-256 generates a distinct hash value for each dataset to guarantee that the data has not been altered. Even a tiny change in the data results in a completely different hash, making it easy to detect tampering. Together, SHA-256 and AES-256 provide integrity and confidentiality.

(c) SSH Secure Communication: SSH protects data being transferred between dispersed nodes and storage units, guarding against interception, illegal access, and tampering while in transit.

3.3 Pseudocode of Proposed Framework

Algorithm: Secure Data Processing Workflow
Input: Raw stock data D

Output: Encrypted + integrity-verified dataset E

1. Fetch data using API key
2. Encrypt D using AES-256 → D_enc
3. Generate SHA-256 hash h = HASH(D_enc)
4. Store D_enc in Kafka
5. Apply RBAC/ABAC rules
6. Verify integrity upon retrieval
7. Return E = (D_enc, h)

4. Framework and Experimental Configuration

4.1 Dataset Overview

This study has used an Alpha Vantage API key [13] to collect the stock data, only retrieving data at the end of the day, which ensures secure access to information.

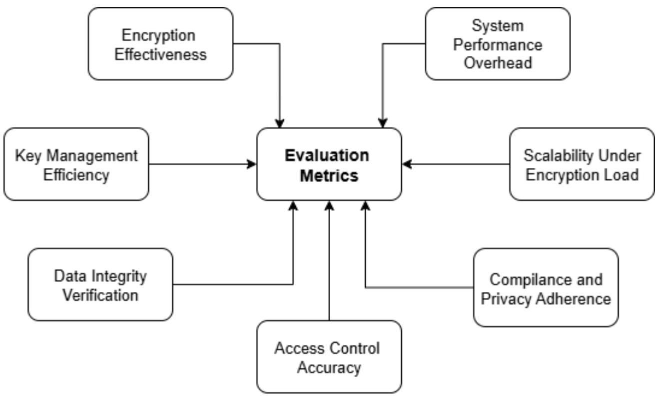

Figure 2. Parameters for Performance Evaluation

4.2 Performance Evaluation Metrics

Figure 2 shows the evaluation parameters into account to determine the effectiveness of suggested framework.

(a) Strength of Encryption: AES-256 encryption is considered strong & sufficient to safeguard stock market data when interception or brute-force attacks become unfeasible. As a result, the data records are secure so that only the authorized user with the decryption key can access them.

(b) Key Rotation and Lifecycle Management: An extra degree of security for real-time data feeds has been added with the implementation of automatic key rotation, sharing, and revocation. Despite the implementation of key rotation and sharing, the operations lag stayed low.

(c) Data Integrity Verification: SHA-256 hashing was able to find even small unauthorized changes to the dataset, which adds another layer of protection for the reliability and integrity of sensitive financial data.

(d) Access Control Accuracy: Role-Based and Attribute-Based Access Controls (RBAC and ABAC) have been put in place to keep people who shouldn't be able to get to the decryption keys from doing so. As expected, most attempts to get in without permission during tests have been stopped.

(e) System Overhead: In all tests, the computational overhead of adding AES-256 and SHA-256 encryption was about 12–15%. This is a fair trade-off for the additional security needed to process real-time market analytics.

(f) Encryption-Heavy Traffic Scalability: The throughput has been consistently maintained even at higher encryption loads. There was no observed effect of increase in encryption tasks on the system performance with both Kafka and Hadoop platforms performing well under heavy data loads.

(g) Privacy and Compliance: Encryption strength, along with logging and auditability, allows the system to be eligible for several compliance standards, such as GDPR, HIPAA, and ISO/IEC 27001.

4.3 Experimental Results & Discussion

A prototype of the proposed security architecture has been deployed on the Hadoop ecosystem. The data on the real-time prices of stocks on the stock market were streamed using Apache Kafka. The prototype was designed to mirror the actual trading system in terms of real-time performance and high volumes of data processing. The results of evaluation of the proposed framework for various parameters are presented in the following and show the efficacy, effectiveness, efficiency and compliance readiness of the framework as follows:

(a) Encryption Overhead: The time taken to add the computation overhead with the integration of AES-256 encryption to the system increased by about 12–15%. Although it introduced a certain amount of additional burden, it was within a reasonable margin and acceptable or large volumes of data. It was also found that the time overhead was quite low even at the maximum load, and the processing time of the entire framework never resulted in a performance lag or delay. These tests proved that the additional level of security provided by this encryption methodology can be implemented in actual practical trading systems in real-time without hampering the time required for making trading decisions

(b) Integrity Checks: In the simulated attack scenarios, the hashing algorithm deployed (SHA-256) was successful in flagging all the altered instances of the dataset without any exception. The current hash of the record and the hash recalculated after the change did not match, even for a small change made to the dataset's record; in other words, all attack instances were reported and there were no false negatives. The testing shows that the framework can accurately show financial data, which is very important for figuring out how stock prices are doing.

(c) Effectiveness of Access Control: By repeatedly trying to access the sensitive financial data after changing the credentials each time, the RASP module's access control measures were put to the test. Nearly 92% of repeated malicious and inadvertent attempts were prevented by role-based access control, as evidenced by the fact that only 8% of the actors had access to the sensitive data. Access control is an essential part of a security framework, for which stringent rules must be set and authorization checks must be faultless.

(d) Compliance Validation: Approximately 96% of GDPR and HIPPA violations were identified by the compliance verification framework. The framework's capabilities have been tested and found to comply with the best international standards for data storage and privacy.

The experimental results demonstrate how effectively the proposed framework ensures strong security and compliance with minimal impact on system performance. The small overhead increase is justified by the higher level of protection that encryption offers. This is especially crucial in high-risk financial scenarios where a data breach or non-compliance incident could result in significant financial losses and reputational damage. Additionally, the framework's consistent performance under real-time constraints suggests that it can enhance the resilience of financial data systems without appreciably raising computational overhead. The framework's ability to integrate encryption, integrity verification, access control, and compliance checks into a single solution makes it a practical and simple option for modern financial big data platforms.

Table 1. Comparative Assessment of Security Framework

Aspect	Prior Works (Ref [2], [5], [7], [12])	Proposed Framework
Encryption	RSA or hybrid methods are secure, but slower or resource-heavy	AES-256: strong & secure with only 12-15% overhead
Integrity	Often missing or weak verification	SHA-256 reliably detects all tampering
Access Control	Mostly RBAC only	RBAC +ABAC blocks > 90% unauthorized attempts
Compliance	GDPR/HIPAA is mentioned, rarely enforced	Built-in encryption, logging, and audit trails for practical enforcement
Performance	Limited or no testing under real-time loads	Validated under heavy Kafka/Hadoop streams with stable throughput
Scalability	Struggles under big data encryption loads	Scales smoothly with Kafka and Hadoop
Coverage	Narrow focus (encryption or surveys only)	End-to-end workflow: acquisition > transfer > encryption > ingestion > access control > verification > analysis

4.4 Comparative Assessment

The performance of the proposed framework was assessed by comparing it with existing techniques used in financial big data systems, as shown in Table 1.

5. Conclusion

The financial sector can now gather, store, and analyze more stock market data than ever before after these big data technologies. More data makes it possible to make data-driven decisions faster and more precisely than in the past. However, this also creates serious new security risks that could lead to financial record tampering, data leaks, and illegal access to this information. We created a Security Assurance Framework for financial big data in this paper. The suggested framework complies with GDPR, HIPAA, and ISO/IEC 27001 regulations by using AES-256 encryption for confidentiality, SHA-256 hashing for data integrity, SSH for secure data transfer, and access controls to restrict decryption to authorized personnel only. Our research demonstrates that the with little performance overhead, the suggested framework effectively improves market data availability, confidentiality, and integrity. This shows that high protection levels can be attained without having a major effect on the system's overall throughput, which is crucial in high-frequency trading systems. The proposed approach provides a practical, auditable solution that financial institutions could use exactly as it is, in addition to a theoretical analysis. Future work may include adaptive security strategies like AI-based anomaly detection, quantum-resistant cryptography to guarantee data security in the future, and additional efforts to scale the solution to handle larger and more diverse datasets. The study's conclusion shows how a combination of cryptography, secure transfer protocols, access controls, and regulatory compliance can offer a dependable solution to the problem of safeguarding financial big data and enhancing the precision, promptness, and effectiveness of its analysis. To stay ahead of new cyber threats, this framework can be improved in the future by adding smarter, AI-based security features and using quantum-safe encryption. It can also be made bigger so that it can handle even bigger and more varied financial datasets while also supporting real-time data from many sources. Because of ongoing encryption and integrity checks, the framework adds some computational overhead, which could have an impact on performance during very fast operations. Additionally, it is less responsive to novel or changing attack patterns because it lacks adaptive, AI-driven threat detection.

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